

U.S. Health Survey SQL Project

Blood pressure, body size, and physical activity patterns

Data source: NHANES 2021–2023 · Tools: Python, SQLite, SQL, Pandas, Matplotlib

Project goal

Build a relational SQL database from raw health survey components and use SQL + visualization to answer two focused questions about blood pressure and activity.

Main message: age and sex show clear BP differences; high-BP participants have heterogeneous activity levels.

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Data model: one participant key links four health tables

This design prevents broken joins after separately cleaning each survey component.

Cleaned tables

participants: 11,933 rows

demographics: 6,488 rows
physical_activity: 8,057 rows
body_measurements: 6,749 rows
blood_pressure: 7,518 rows

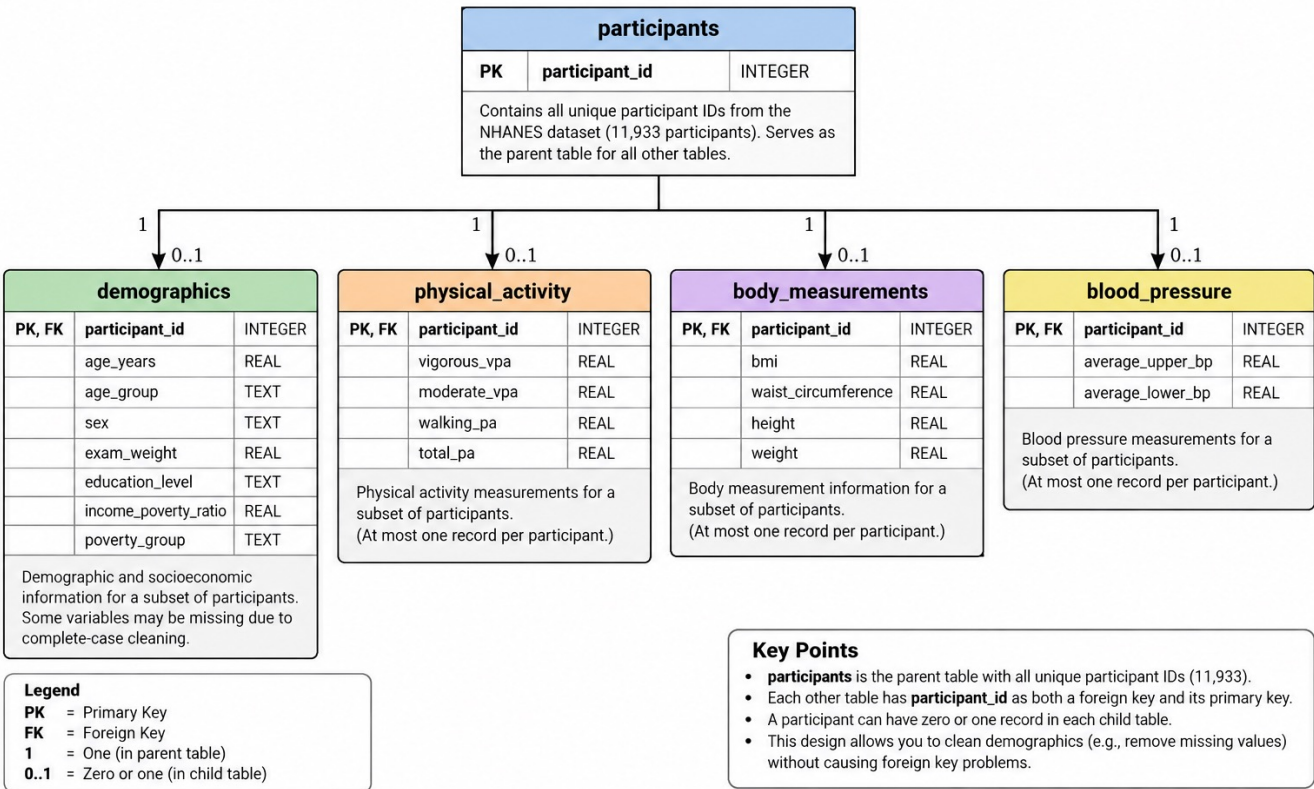
Each child table can have fewer rows because each component has its own missingness and cleaning decisions.

Design decision

I used a separate participants table as the parent key table. Each analysis then joins only the tables needed for the question.

Entity-Relationship Diagram (ERD)

NHANES Database with Separate Participant Key Table



Primary key: participant_id in participants
Foreign key: participant_id in each health table

Preparing analysis variables

I created comparable groups and weekly measures before running the SQL analysis.

Age categories

Young adults: 20–39
Middle-aged adults: 40–59
Older adults: 60+

Cleaned demographics counts:
Young: 1,733
Middle-aged: 1,854
Older: 2,901

Blood pressure

Repeated measurements were averaged.

`average_upper_bp` = mean upper BP
`average_lower_bp` = mean lower BP

Upper BP is the “top number” in a BP reading.

Activity conversion

$\text{minutes/week} = \text{frequency} \times \text{minutes} \times \text{unit factor}$

Day = 7
Week = 1
Month $\approx 12/52$
Year $\approx 1/52$

Example: 3 times/week $\times$ 45 minutes = 135 moderate minutes/week

If frequency = 0, weekly minutes were set to 0 so inactive participants were not removed by mistake.
This standardization made activity comparable across participants before joining tables in SQL.

Two research questions drive the analysis

Each question uses SQL joins and aggregated or filtered results pulled into Pandas.

Research question 1

How does average blood pressure differ across age groups and sex?

Hypothesis: older adults might show higher average upper blood pressure.

SQL joins: demographics + blood_pressure

Visuals: grouped bar charts

Research question 2

Among participants with upper blood pressure above 130 mmHg, what are their moderate and vigorous physical-activity levels?

Hypothesis: high-BP participants would mostly show lower weekly activity.

SQL joins: blood_pressure + physical_activity + demographics

Visuals: scatter plots by age group

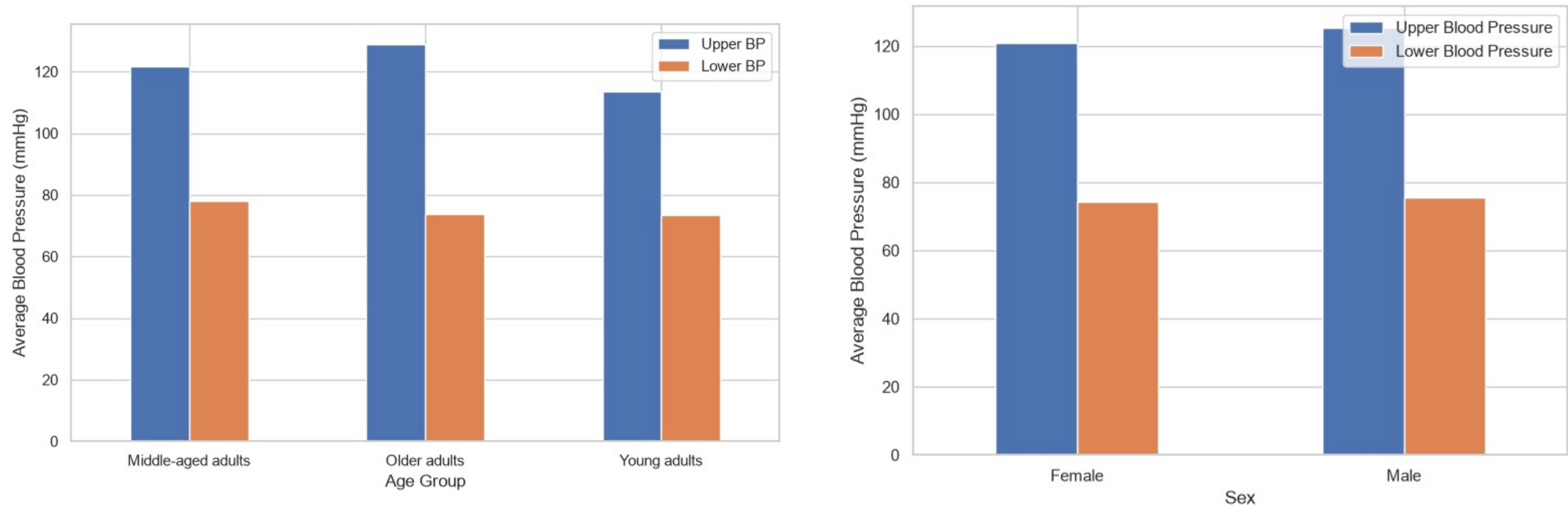
Supporting view

BMI is used as an exploratory context variable, not as a causal explanation. It helps check whether body size patterns overlap with high upper BP.

SQL answers the data question → Pandas receives the result → charts communicate the finding

RQ1: Older adults show the highest average upper blood pressure

SQL grouped participants by age group and sex after joining demographics with blood pressure.



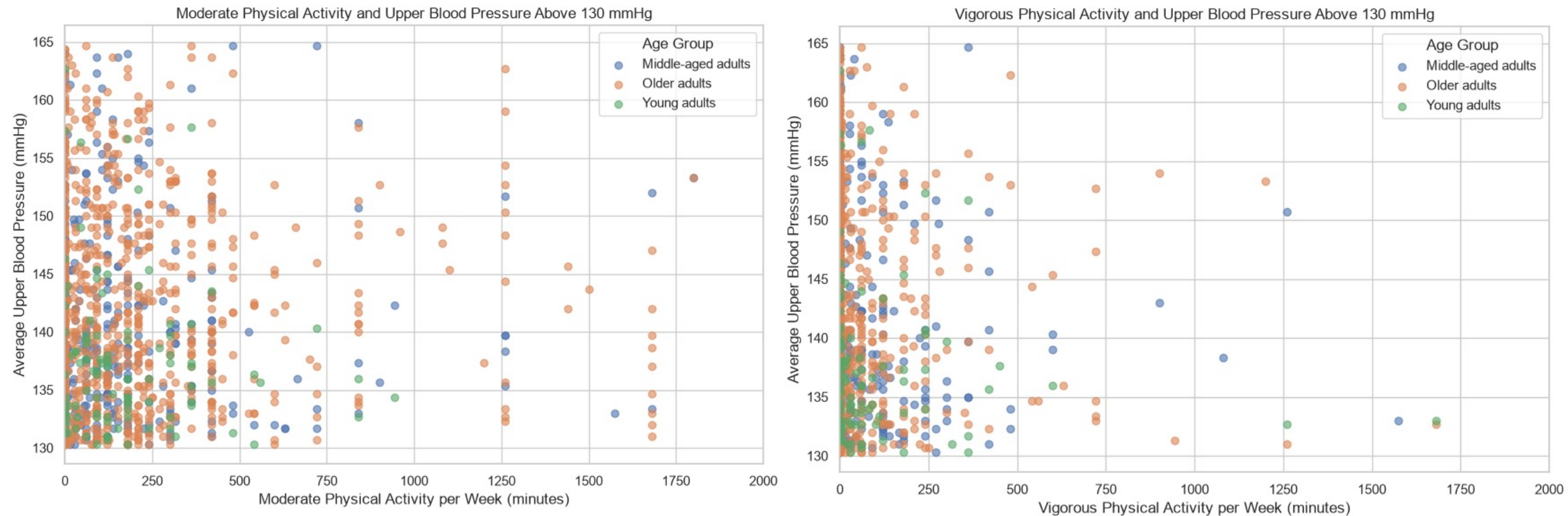
Finding

Upper BP increases clearly from young to older adults. Males show slightly higher average upper and lower BP than females.

Verdict: hypothesis supported for age; partially supported for sex.

RQ2: High-BP participants have diverse activity levels

I filtered participants with upper BP > 130 mmHg and visualized weekly activity by age group.



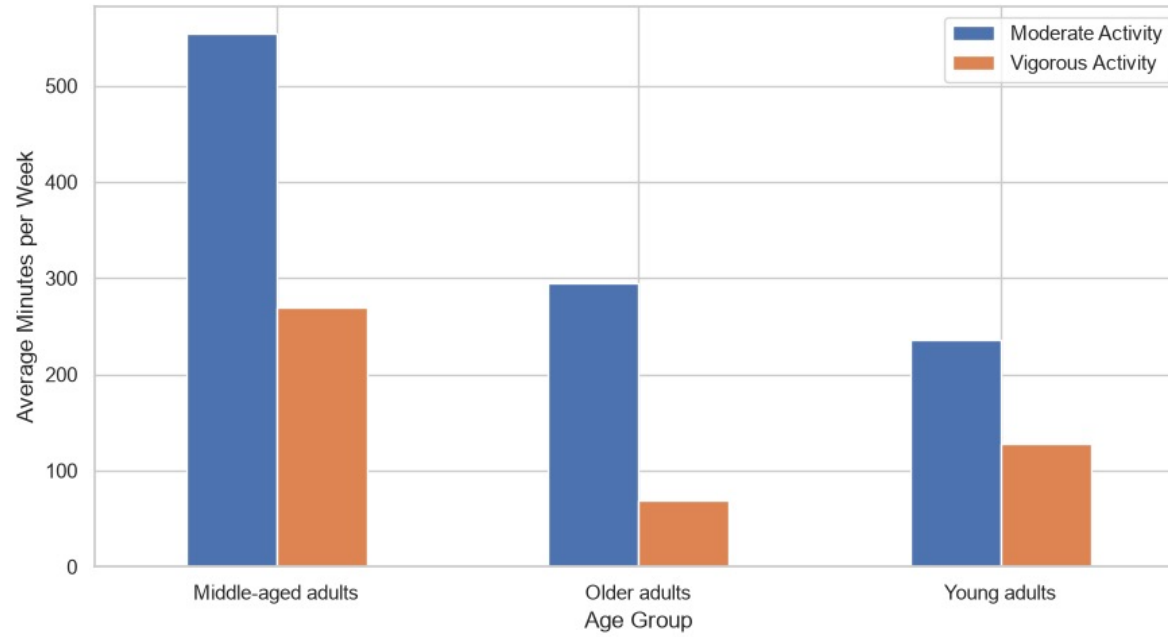
Finding

Many high-BP participants cluster at lower activity levels, especially for vigorous activity. However, some high-BP participants report high weekly activity.

Verdict: hypothesis partially supported — low activity is common, but the pattern is heterogeneous.

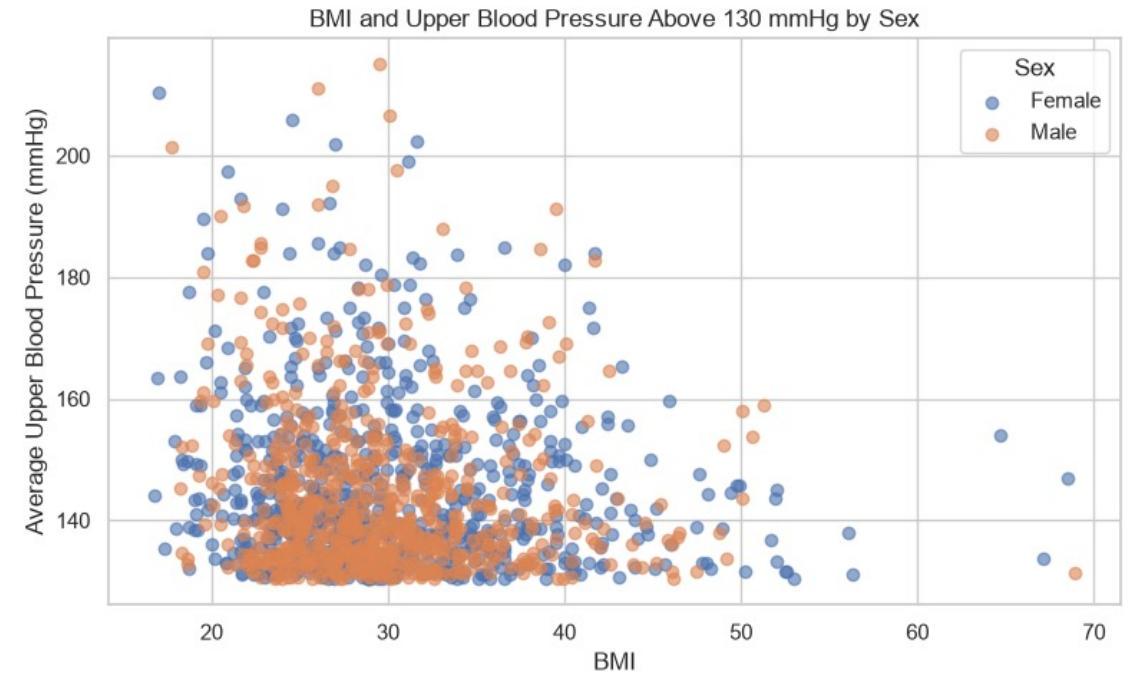
Supporting context: activity and BMI patterns

These charts help interpret the main results but do not prove causality.



Activity by age

Middle-aged adults show the highest average weekly activity in this cleaned sample. Older adults show lower vigorous activity.



BMI + BP

BMI and upper BP above 130 overlap substantially by sex. There is no simple visual separation.

Use these as context, not as causal explanations.

Conclusions, limitations, and next steps

The project shows descriptive patterns and where more rigorous modelling is needed.

Main conclusions

1. Age is the clearest BP pattern: older adults have the highest average upper BP.
2. Males show slightly higher average BP than females.
3. Among participants with upper BP > 130, low activity is common but not universal.

Limitations

- Physical activity was self-reported, so participants may have over- or underestimated how active they were.
- The analysis shows patterns and associations, but it cannot tell us whether physical activity directly caused the differences in blood pressure.

Next steps

- Include other health and lifestyle factors:
 - smoking, diet, medication use, sleep, and existing health conditions
- to better understand differences in blood pressure.

Final message: SQL and visualization revealed clear age-related BP patterns, but activity among high-BP participants is mixed — a useful result, not a failure.